

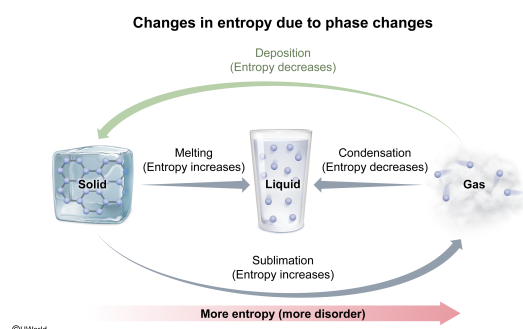
Which of the following processes will result in the greatest decrease in entropy in a sample of a chemical substance?

- ✓ ☒ A. Deposition [73%]
☐ B. Melting [2%]
☐ C. Condensation [10%]
☐ D. Sublimation [13%]

Correct

73% answered correctly

Explanation:



Matter commonly exists in **three states (phases)**: solid, liquid, or gas. The **characteristics of the particles** (ie, atoms, molecules) in each phase determines the relative amount of **entropy** of each phase:

- **Solids** have constituent particles that are **closely packed** and held in a **fixed arrangement** by strong mutual attractions that prevent translational motion (ie, movement from one place to another) within the solid. As such, solids have the **lowest entropy** of the three common phases.
- **Liquids** have particles that are held **close together** by significant mutual attractions, but the particles are **not fixed in place** and can move around each other into different arrangements (ie, translational motion occurs). Consequently, liquids have **more entropy than solids**.
- **Gases** have particles that are **spaced far apart** due to minimal attractive forces. As a result, the particles **move freely and disburse** into continuously changing arrangements within their container. Accordingly, gases have the **highest entropy** of the three common phases.

During deposition, a substance in the gas phase transitions directly to the solid phase (ie, the substance goes from the phase with the highest entropy to the phase with the lowest entropy). Therefore, among the given processes, deposition will result in the greatest decrease in entropy.

(Choice B) **Melting** (ie, going from the solid phase to the liquid phase) *increases* the entropy of a substance.

(Choice C) Condensation (ie, going from the gas phase to the liquid phase) does decrease the entropy of a substance, but condensation yields a smaller decrease in entropy than that resulting from deposition.

(Choice D) Sublimation (ie, going from the solid phase directly to the gas phase) is the opposite of deposition and results in the *largest increase* in entropy among the given processes.

Educational objective:

Solids have closely packed particles held in a fixed arrangement. Liquids have particles in close contact that can move around each other in various arrangements. Gases have particles that are spaced far apart and can move freely into numerous arrangements. As such, solids have lower entropy than liquids, and liquids have lower entropy than gases.

Time spent:0 sec

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General Chemistry

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Thermodynamics, Kinetics & Gas Laws